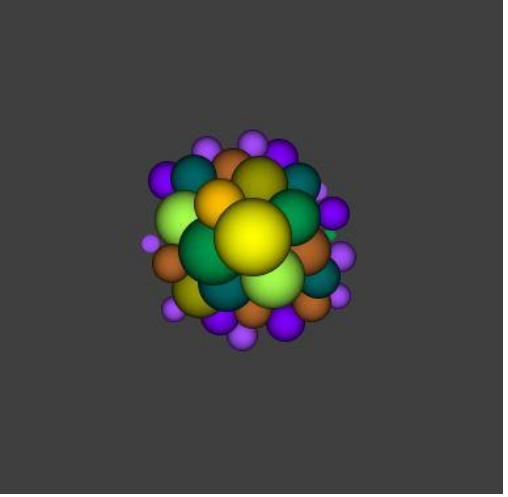
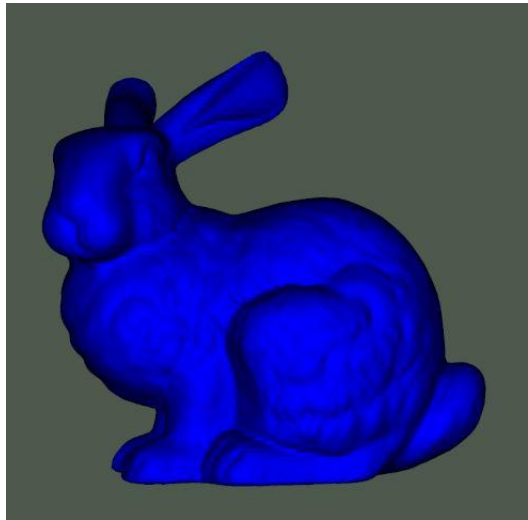


CS 440: Computer Graphics

Week 5

Lecture 9: Affine Transformations



Acknowledgements: Cem Yuksel, University of Utah; Waqar Saleem, Habib University

Announcements and Recap

- Assignment due Friday
- Syllabus Updates

- Recap:
  - 3D Primitives
  - Coordinate System
  - Change in Coordinate System
  - Coordinate Frame
  - Change in Coordinate Frame
  - Homogenous Coordinates

Today: Affine Transformations

- Affine Transformations
  - Transformation Matrix
  - Translation
  - Rotation
  - Scaling
  - Shear
- Concatenating Transformations
- General Transformations
- Model Instance Paradigm & Scene Graphs

Recall: Homogenous Coordinates

$$\vec{w} = \mathbf{a}^\top \mathbf{v} = \mathbf{b}^\top \mathbf{u} = \mathbf{b}^\top \mathbf{M}\mathbf{v}$$

$$\mathbf{a} = \mathbf{M}^\top \mathbf{b}$$

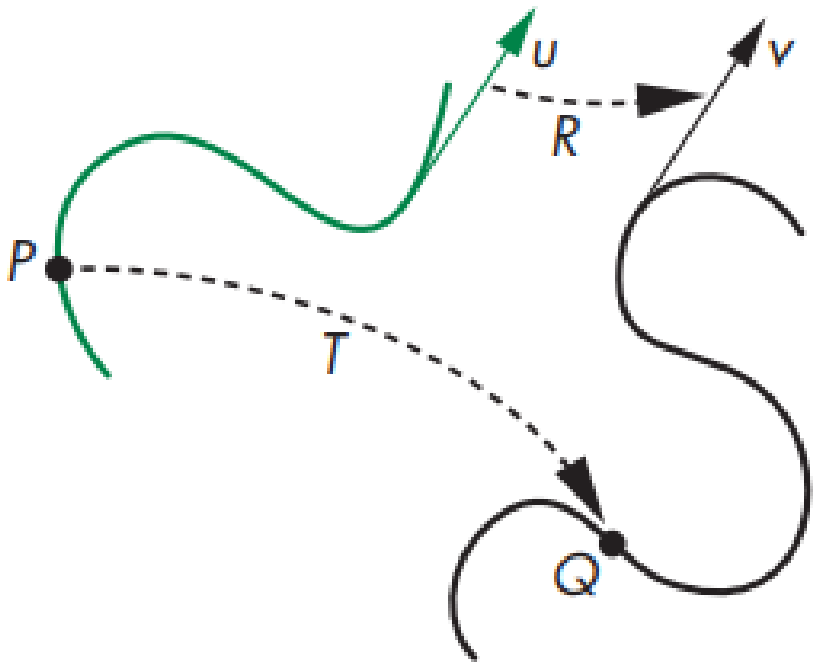
$$\mathbf{M}^\top = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} & \alpha_{14} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} & \alpha_{24} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} & \alpha_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Affine Transformations

- A transformation is a function that takes a point (or vector) and maps it into another point (or vector).

$$Q = T(P)$$
$$\vec{v} = R(\vec{u})$$

$$Q = f(P)$$
$$\vec{v} = f(\vec{u})$$



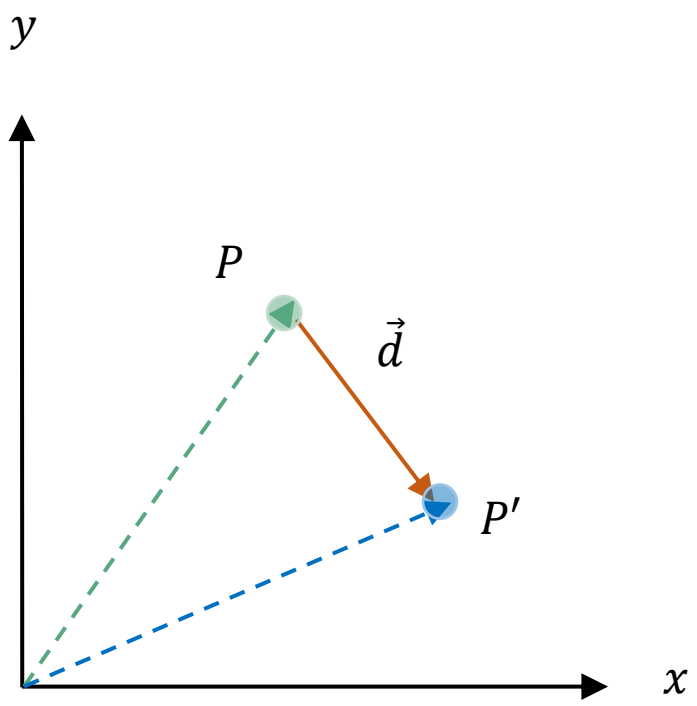
Affine Transformation

$$f(\alpha P + \beta Q) = \alpha f(P) + \beta f(Q)$$

Translation

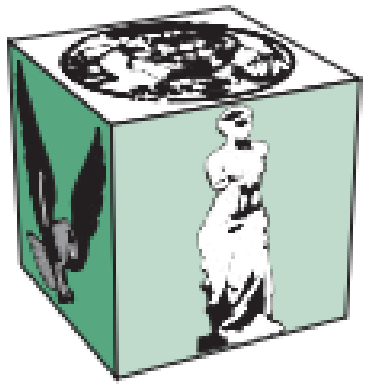
- Displacement of points by a vector

$$P' = P + \vec{d}$$



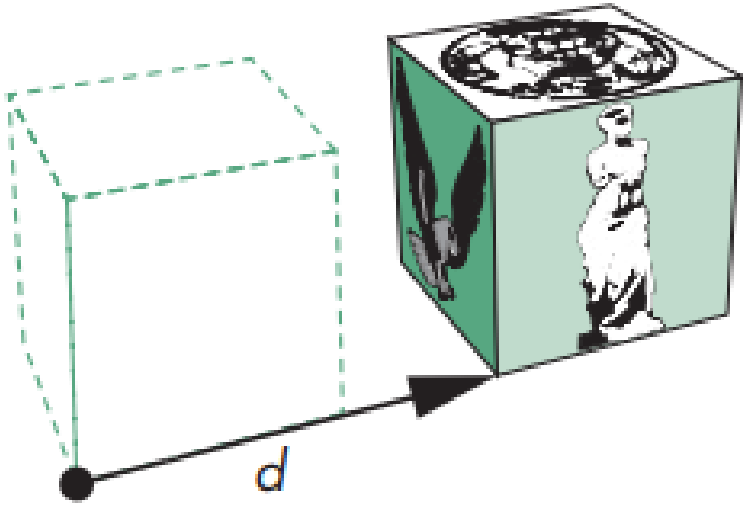
Translation

- Displacement of points by a vector



$$P' = P + \vec{d}$$

$$P = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \qquad P' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}, \qquad \vec{d} = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix}$$



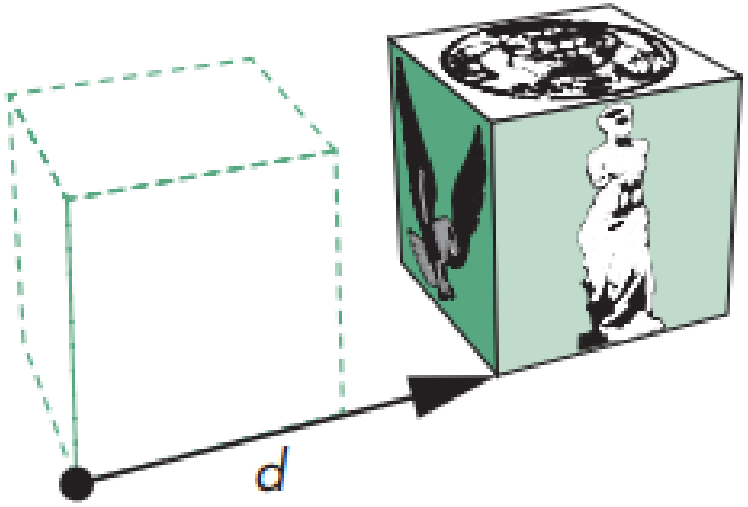
Translation

$$x' = x + d_x$$

$$y' = y + d_y$$

$$z' = z + d_z$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x + d_x \\ y + d_y \\ z + d_z \end{bmatrix}$$

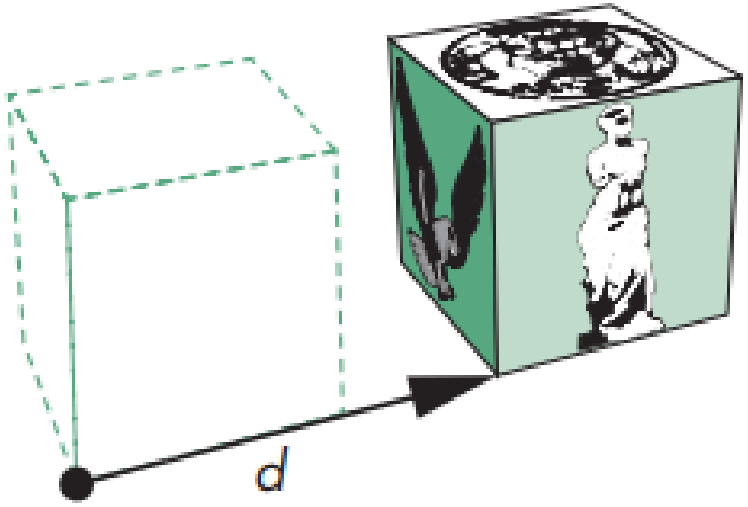


Translation

$$P' = TP$$

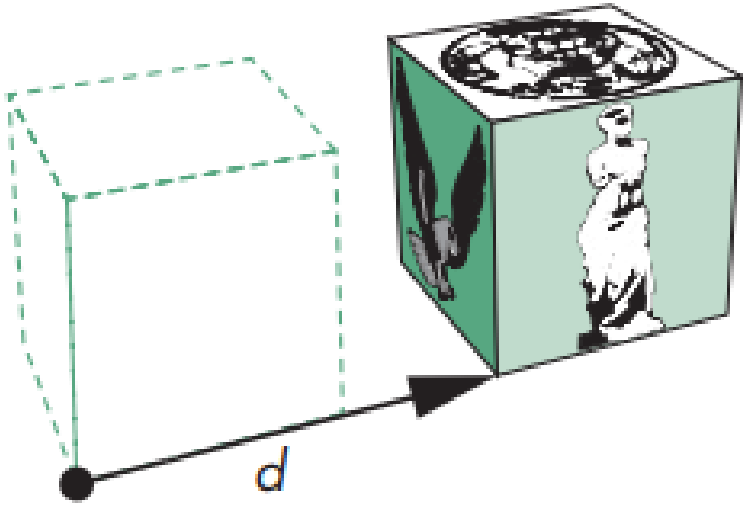
$$P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}, \qquad P' = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}, \qquad \vec{d} = \begin{bmatrix} d_x \\ d_y \\ d_z \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Translation

$$T = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

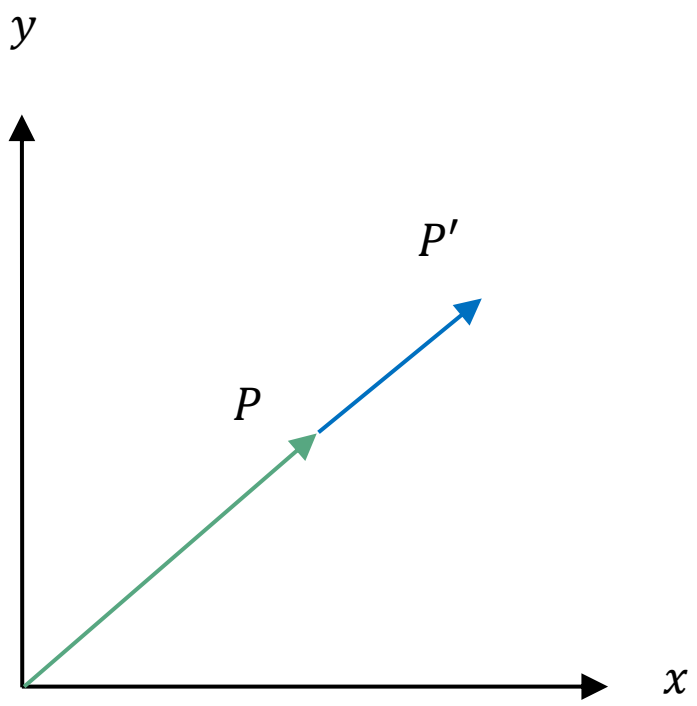


- Inverse Translation Matrix?

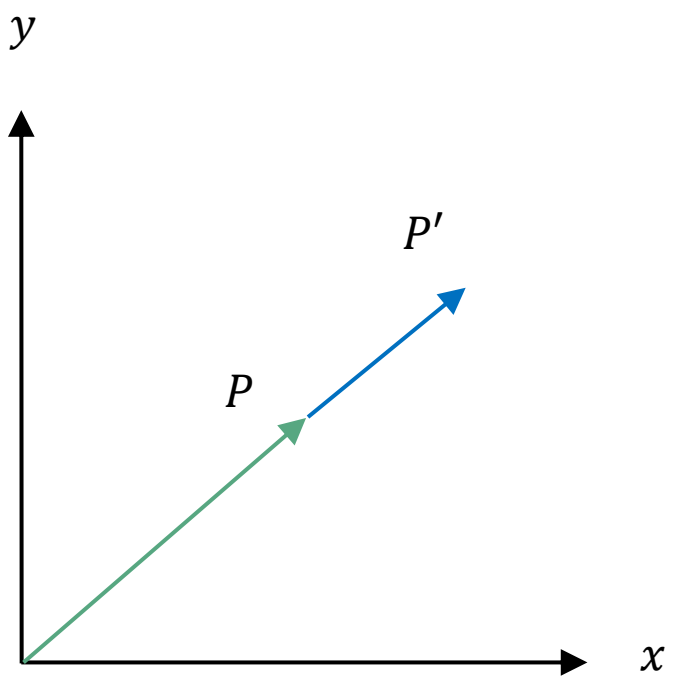
Scale

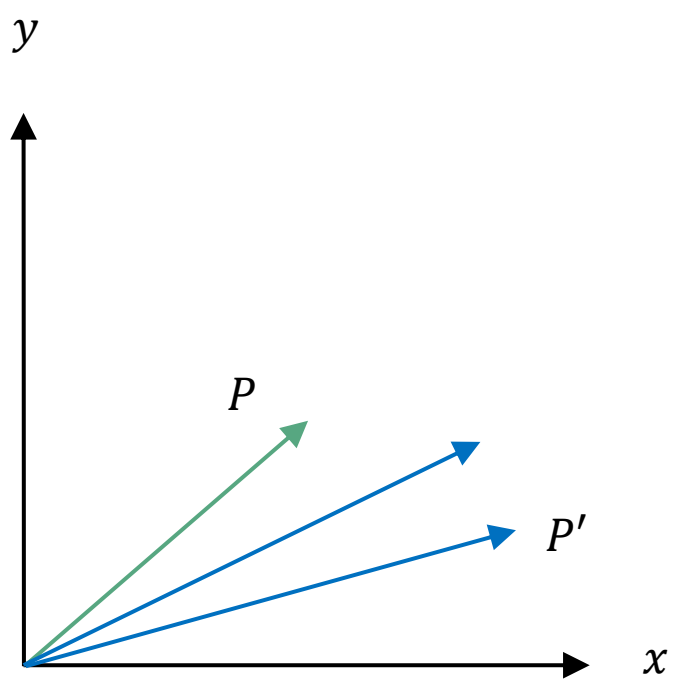
- Making an object bigger or smaller by a factor
- Scaling of a point/vector

$$P' = SP$$

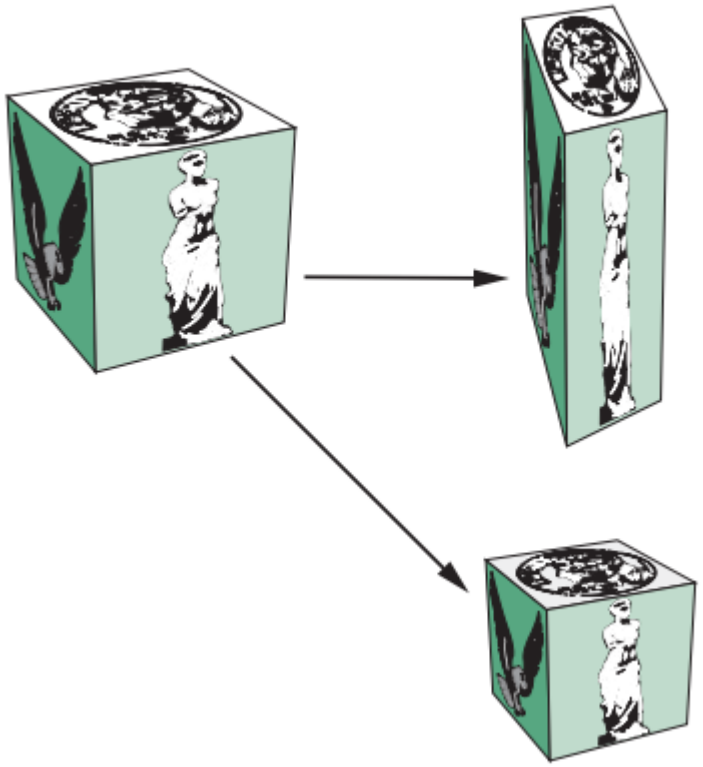


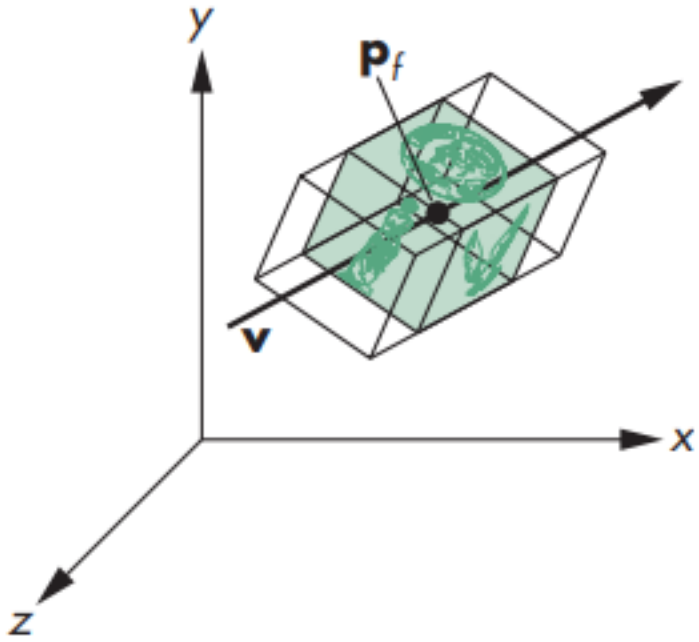
Uniform vs Non-uniform Scaling





Scaling





Scaling

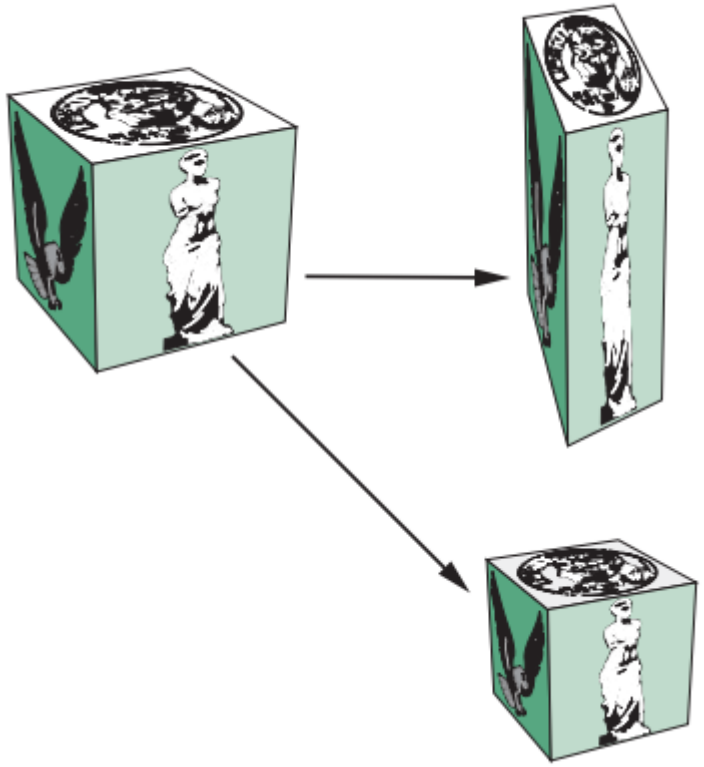
$$P' = SP$$

$$x' = S_x\, x$$

$$y' = S_y\, y$$

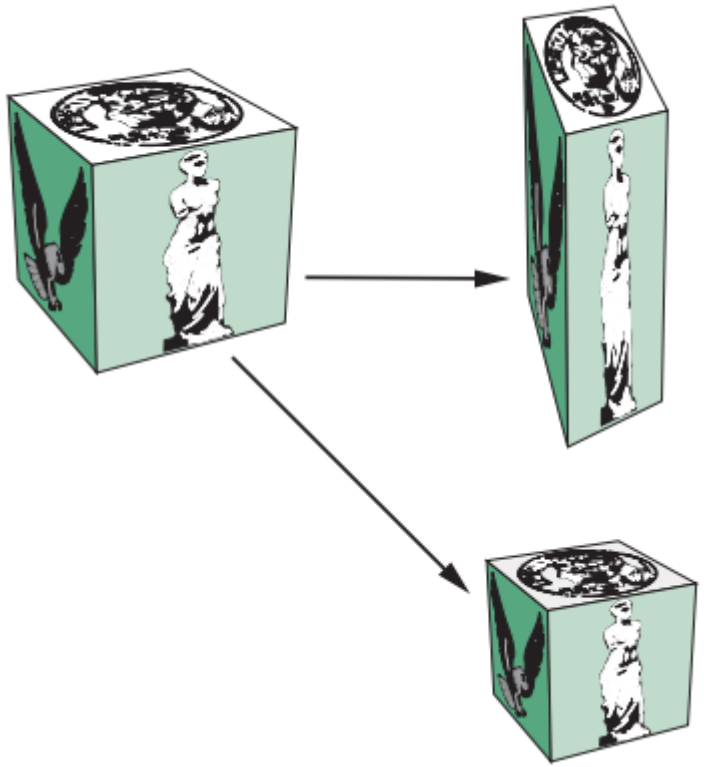
$$z' = S_z\, z$$

$$P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}, \qquad P' = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$



Scaling

$$S = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



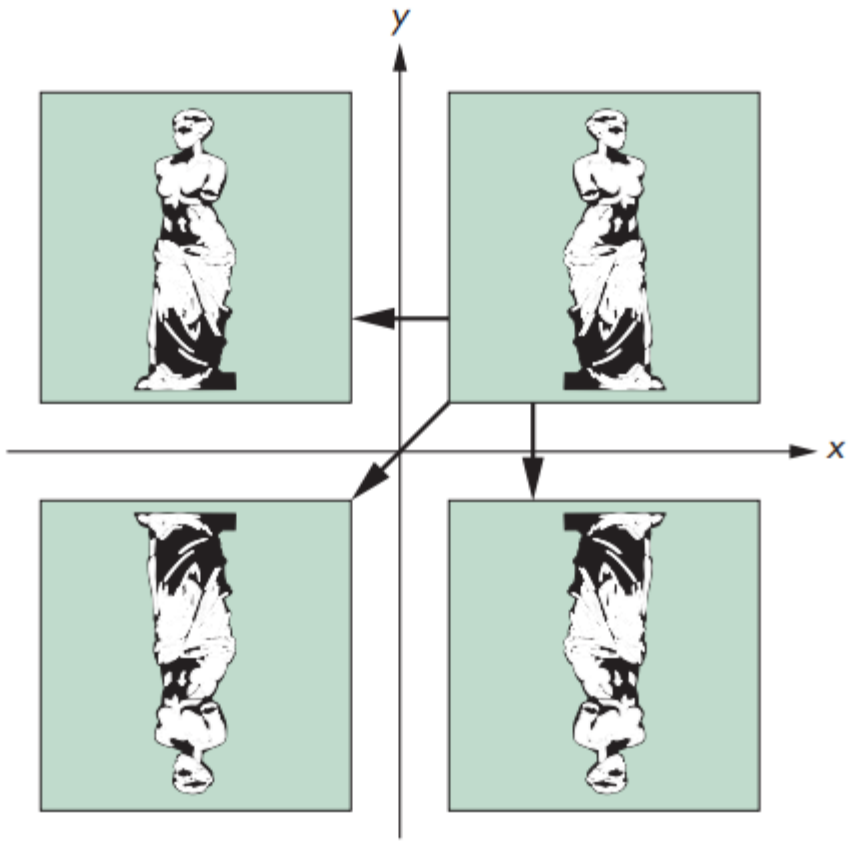
- Inverse Scaling Matrix?

Reflection

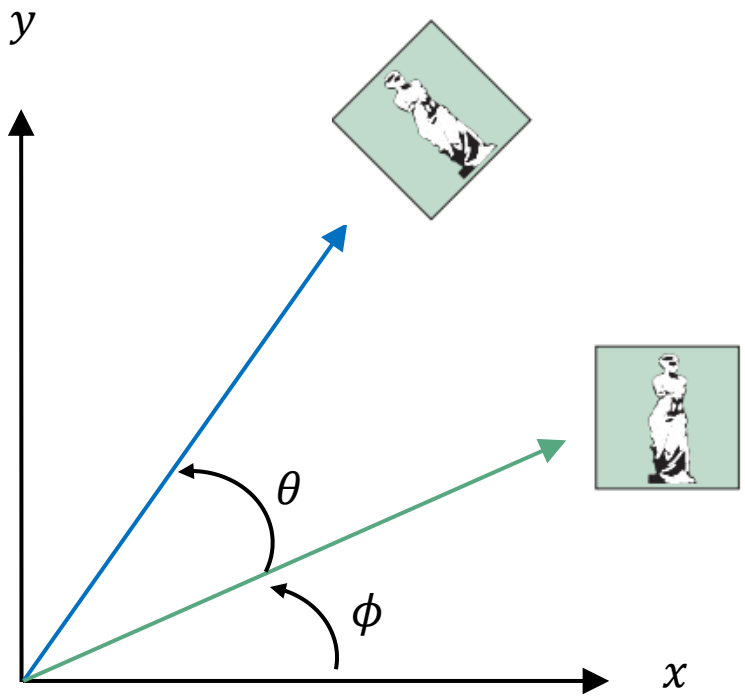
- Special case of scaling
- What would be its transformation matrix?

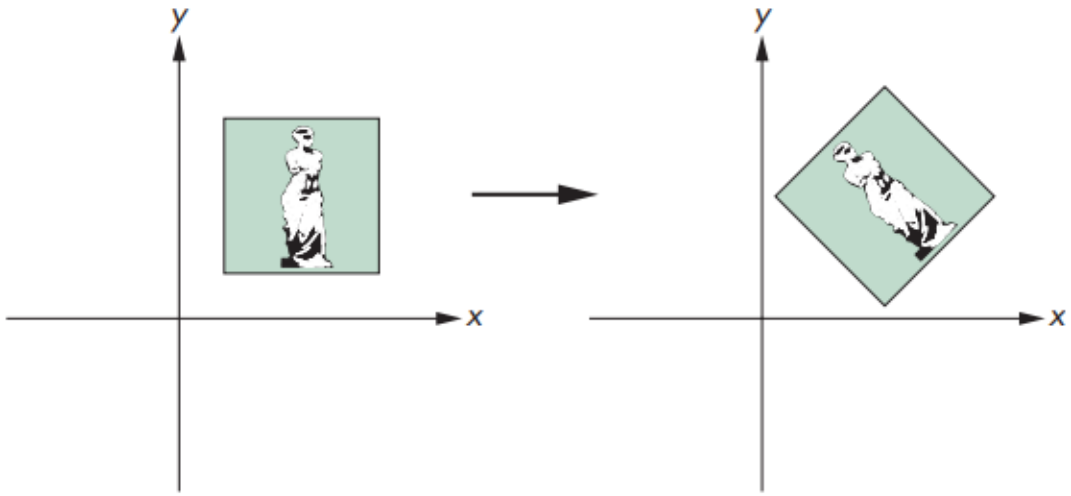
- Reflection along XY:

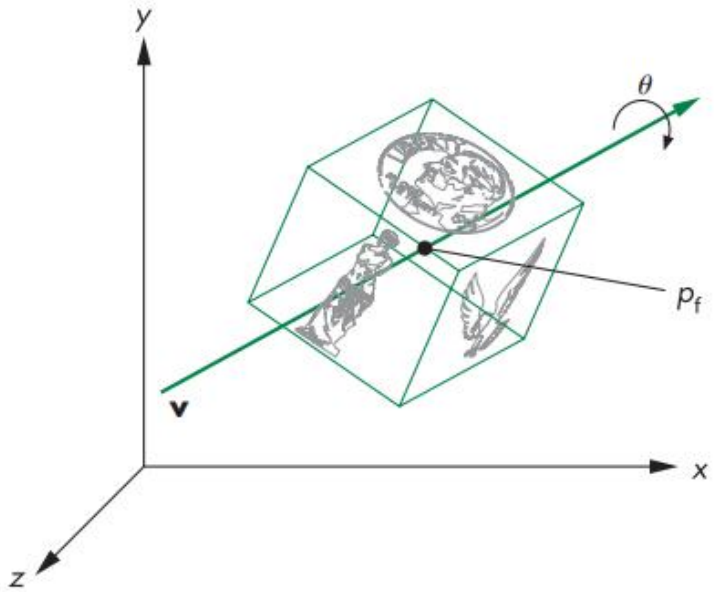
$$R_{xy} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Rotation





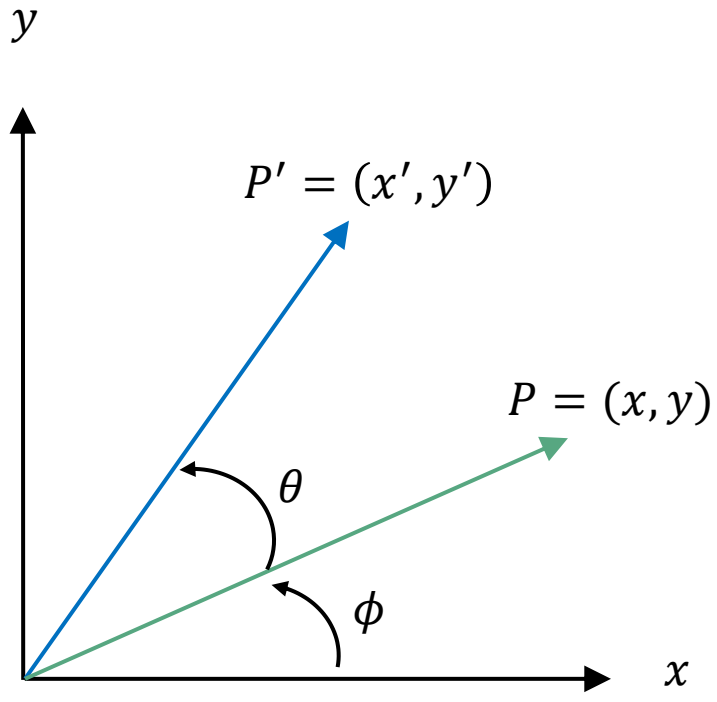


Rotation

- Rotation along a fixed point (or origin) w.r.t an angle
- In polar form (or through trigonometry):

$$x = \rho \cos \phi$$
$$y = \rho \sin \phi$$

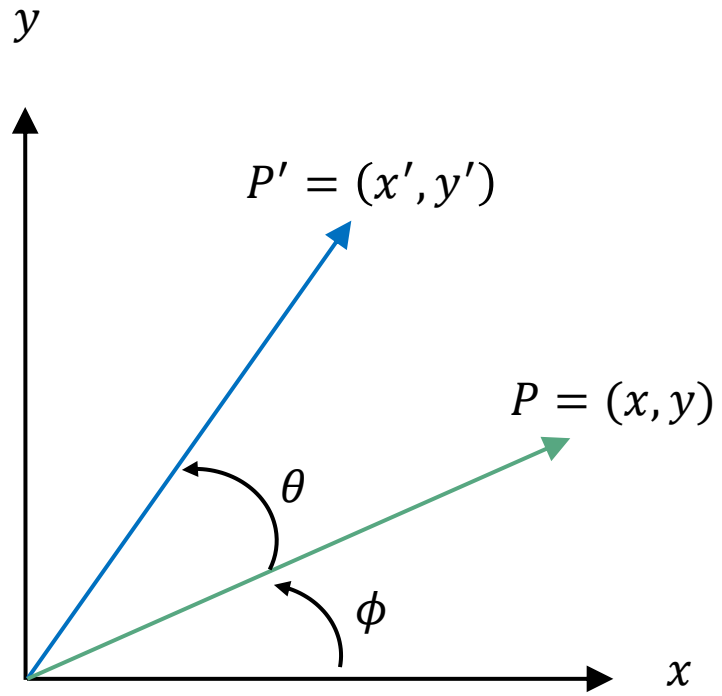
$$x' = \rho \cos(\theta + \phi)$$
$$y' = \rho \sin(\theta + \phi)$$



Rotation

$$x' = \rho\cos\phi\cos\theta - \rho\sin\phi\sin\theta$$
$$x' = x\cos\theta - y\sin\theta$$

$$y' = \rho\cos\phi\sin\theta + \rho\sin\phi\cos\theta$$
$$y' = x\sin\theta + y\cos\theta$$



- In matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

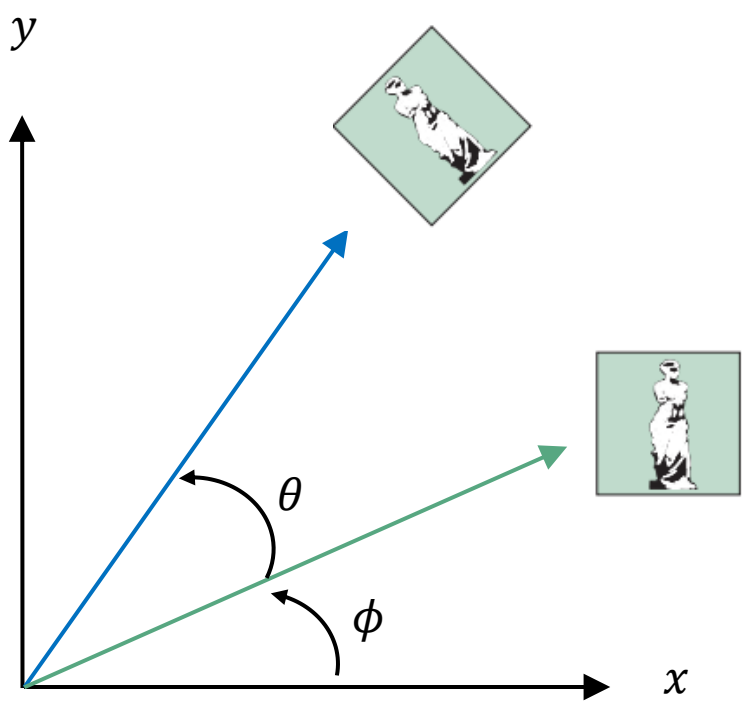
Rotation

- Rotation along different axis is not the same
- Rotation along z-axis:

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$
$$z' = z$$

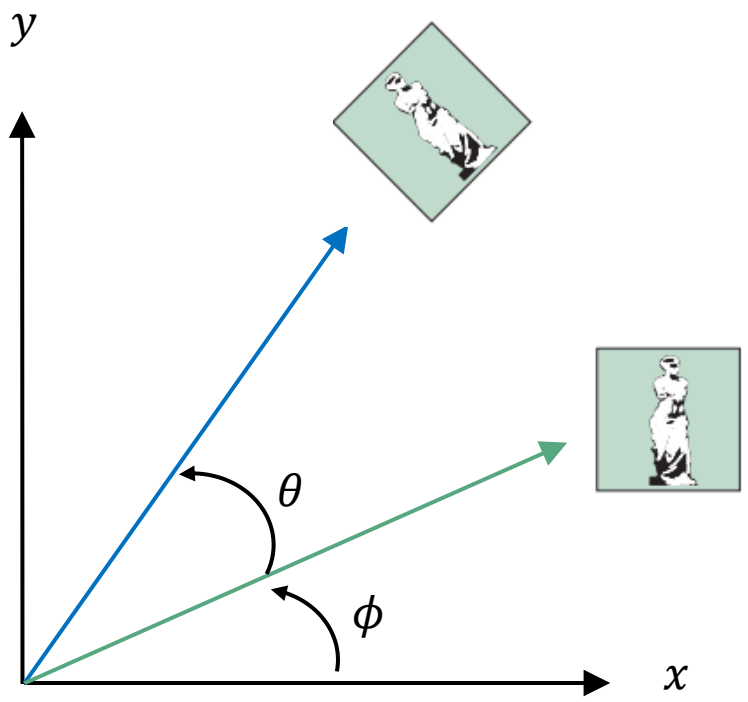
- In matrix form:

$$P' = R_z P$$



Rotation

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



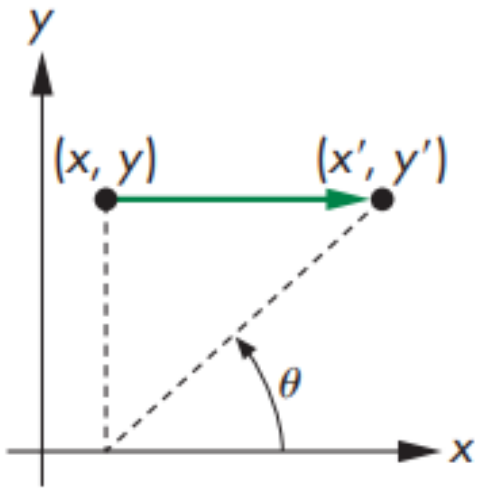
- Rotation along x and y axis?
- Inverse rotation matrix?

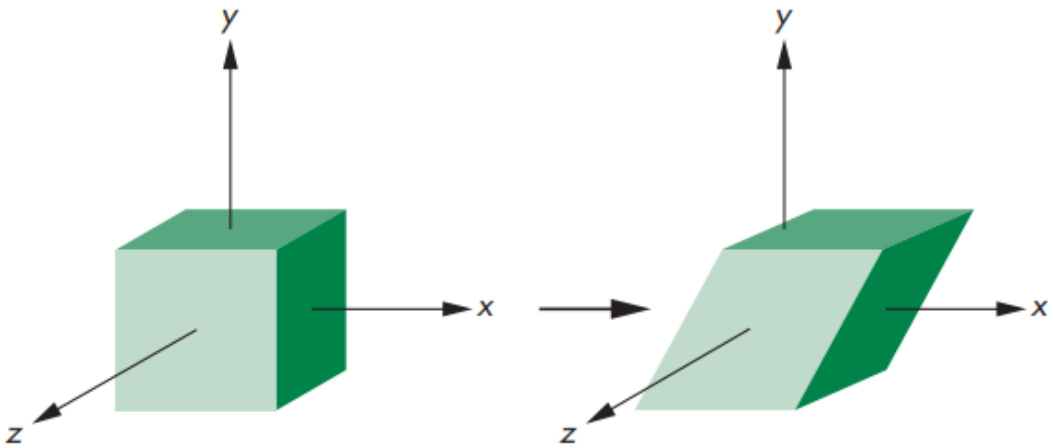
Shear (or "Skew")

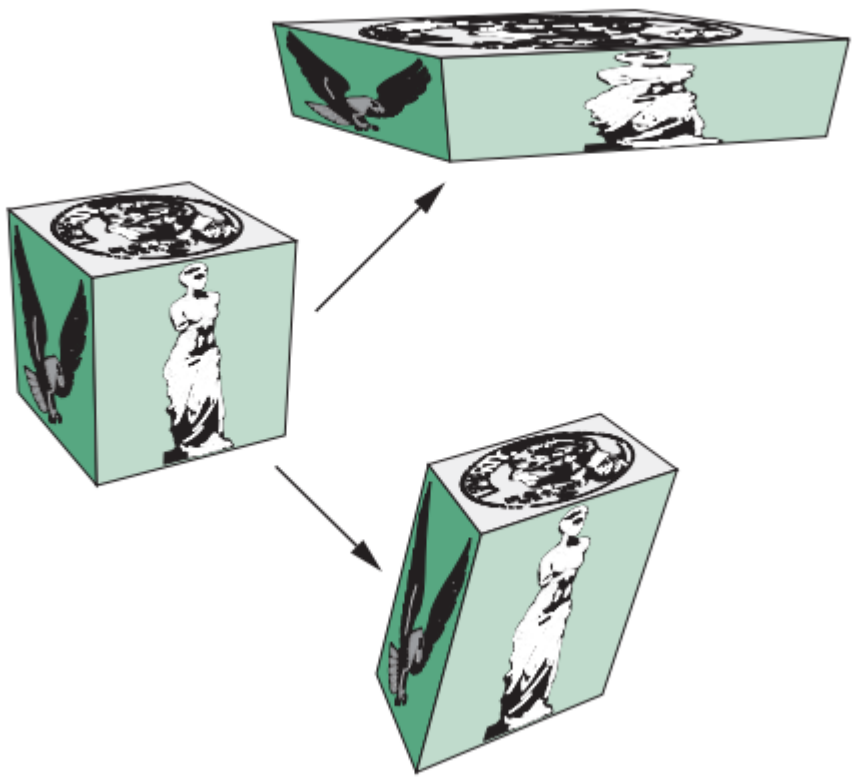
- Distorted and skewed objects

$$x' = x + y\cot\theta$$
$$y' = y$$
$$z' = z$$

$$H_x = \begin{bmatrix} 1 & \cot\theta & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$





Rigid vs Non-Rigid Body Transformations

- Is the shape or volume of an object preserved after transformation?
- Examples?
  - Non-rigid body: Shear, Scaling
  - Rigid body: Translation, Rotation